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An IoT-enabled AI system for real-time crop prediction using soil and weather data in precision agriculture

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ABSTRACT

Context: Precision agriculture leverages advanced technologies such as the Internet of Things (IoT) and artificial intelligence (AI) to enhance crop productivity by providing data-driven insights. In Bangladesh, optimizing crop recommendations using real-time soil and environmental data is crucial for improving agricultural decision-making. However, integrating AI models with IoT devices for instantaneous crop prediction remains a challenge due to computational constraints and the need for model interpretability.

Objective: This study aims to develop an IoT-based crop prediction system that utilizes real-time data on soil nutrients, pH, and weather conditions. The system employs machine learning and deep learning techniques to recommend suitable crops based on local environmental factors. The implementation focuses on deploying the best-performing models on an edge device for real-time predictions as well as ensuring accuracy, efficiency, and accessibility for farmers and agricultural stakeholders.

Methods: The system was developed in Bangladesh using proprietary data from the Soil Resource Development Institute, supported by Habiganj Agricultural University, Sylhet. The dataset consists of 3,300 samples covering 22 crops and eight soil and environmental features. Several machine learning algorithms, including Random Forest, Gradient Boosting, and Stacking ensembles, as well as deep learning models such as TabNet, were evaluated for crop prediction. The best-performing models were deployed on a Raspberry Pi 5 edge device for real-time inference. A weather API was integrated for local humidity and rainfall data, while an RS485 7-in-1 agricultural soil sensor provided real-time measurements of nitrogen (N), phosphorus (P), potassium (K), pH, temperature, and soil moisture. The predictions were displayed on the ThingsBoard IoT platform. Model interpretability was enhanced using the explainable AI technique LIME. A user survey involving farmers, agricultural researchers, and students assessed the usability, accuracy, and reliability of the system.

Results and conclusions: The highest accuracy among machine learning models was achieved using Random Forest (95.8%) and Gradient Boosting (95.5%). The Stacking ensemble technique, with Support Vector Classifier (SVC) as the meta-classifier, achieved the highest overall accuracy of 95.9%. Among deep learning models, TabNet performed best with an accuracy of 92%. The Random Forest model was selected for deployment on the Raspberry Pi due to its lowest inference time and compatibility with Python and TensorFlow Lite. User feedback from the survey provided insights into the system's practical effectiveness and potential areas for improvement. The results demonstrate that integrating AI-driven crop recommendation models with IoT devices can support real-time agricultural decision-making, improving precision farming outcomes.

Significance: This study contributes to precision agriculture by demonstrating an IoT-based crop prediction system that integrates AI-driven recommendations with real-time environmental monitoring. The deployment of the best-performing model on an edge device ensures accessibility and efficiency for users in agricultural settings. By leveraging explainable AI techniques, the study enhances model interpretability, fostering trust and usability among farmers and agricultural researchers. The findings highlight the potential of AI and IoT in improving crop

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selection, optimizing resource usage, and supporting sustainable agricultural practices in Bangladesh and beyond. The implementation code and private dataset are available at: https://github.com/SAIFULLAH-SHARAFAT/An-IoT-Enabled-AI-System-for-Real-Time-Crop-Prediction-Using-Soil-and-Weather-Data.

1. Introduction

Bangladesh has an estimated population of 171 million. Primarily as an agronomic nation, its economic activities heavily rely on agriculture, employing approximately 40.6% of the labor force. Nevertheless, the contribution of agriculture to the national GDP has steadily declined in recent years due to several challenges that impede food security, including limited access to advanced technologies, the adverse effects of climate change, natural disasters, poor resource distribution, a decline in arable land, and overpopulation [1].

The rapidly changing climatic conditions in Bangladesh have rendered agricultural endeavors increasingly vital. The country experiences unpredictable monsoon seasons, intermittent droughts, and an increased risk of flooding due to contemporary climate trends. Furthermore, soil composition varies significantly across agroecological zones (AEZs) [2], necessitating the development of targeted methods to enhance fertility tailored to various crops [3]. Combined with resource constraints—such as inefficient irrigation systems and the overuse of chemical fertilizers—traditional farming practices often fail to meet the growing demand for food [4]. In contrast to the counterparts in developed nations, rural agricultural sectors in Bangladesh have yet to reap the benefits of technological advancements.

According to Fig. 1, worldwide IoT endpoints climbed from 3.6 billion in 2015 to 18.6 billion in 2024, while Bangladesh's licensed M2M/IoT SIMs increased from 9.5 million in 2022 to 18.5 million in 2024 [5], illustrating parallel upward trends at vastly different scales.

Modern technological solutions, such as sensors and Internet of Things (IoT) systems, enable monitoring essential crop growth factors, including soil nutrient levels, temperature, humidity, and other critical elements [6]. Farmers can optimize resource allocation and maximize crop yields by utilizing data on such factors. Besides, traditional farming methods in this country are marked by arbitrary crop selection, often disregarding agroecological conditions, soil nutrient profiles, and meteorological data—factors critical to achieving optimal yields. Constrained by limited formal education, many farmers encounter difficulties in effectively managing agricultural resources [7]. As a result, they frequently grow crops that fail to satisfy nutritional demands and forgo essential procedures such as routine soil analysis before fertilizer application or irrigation. These practices contribute to economic strain and suboptimal labor efficiency.

Bangladesh remains vulnerable to the consequences of climate change due to its deltaic landscape and farming systems, which rely heavily on the monsoon. The government has faced the wrath of extreme tropical cyclones, tidal surges, and unpredictable rainfall patterns within a decade, illustrated in Fig. 2. Many examples of storms and tropical cyclones have severely damaged the country's economy. For instance, Cyclone Amphan of 2020 hit Khulna and Mocha in 2023, ruined the crops in Barisal and Remal, and affected Chittagong in 2024. All these storms not only impacted the farming systems but also hindered the poverty-stricken people of Bangladesh, who, without these frequent disasters, are food insecure. There is still hope for agricultural advancements in the country, which will help mitigate the devastation caused by disasters.

Table 1 depicts consolidated national statistics on annual crop production losses attributed to predominant climate-related disasters, specifically, fluvial floods and cyclonic events—spanning the agricultural years 2017–2018 through 2021–2022 [8]. The dataset reveals that catastrophic flooding in 2017 and 2018 resulted in agricultural losses surpassing 900,000 metric tons. Conversely, heightened cyclonic events in 2019 and 2020 caused yield declines approaching 200,000 metric tons. This fluctuation underscores the erratic influence of climate change on agricultural productivity and the consequent socio-economic vulnerability of rural populations in Bangladesh.

Bangladesh has experienced dramatic climatic fluctuations over the past decade, with average temperatures increasing by 0.8^0C and frequently exceeding 37^0C, particularly in the northwestern regions, summarized in Table 2. Rainfall regimes have also become extremely unpredictable, with heavy monsoon occurrences in 2017 and 2021 leading to severe flooding and widespread crop damage. Extreme rainfall can also combine with high humidity levels of 77% to 80%, causing thermal stress and facilitating the propagation of pests and crop diseases. These external stresses often cause yield losses to the agrarian economy, resulting in overstressed cropping calendars and challenges to achieving a food-secure nation.

This work represents modern precision agriculture and offers a practical resource management and crop selection solution. This study proposes a system to optimize farmers' capability to make informed decisions based on real-time soil data and contribute to food security. This system uses essential soil information [9], such as nutrients, soil type, pH, and soil moisture, and articulates the best crop for that land. Machine learning models have been utilized to analyze the macro and micro-nutrients and other information of the soil and weather forecasts efficiently using advanced parallel computing capabilities, suggesting the most suitable crops for specific areas based on previous soil data. This strategy will boost crop yield, reduce pesticide usage and waste, enhance food security, and promote sustainability in the agricultural sector.

The application of the Internet of Things (IoT) [10] in precision agriculture is well established globally; however, its adoption in Bangladesh remains constrained by resource inefficiencies, archaic farming practices, and the absence of data-driven decision support systems. By enabling real-time data collection from sensors, farmers receive actionable insights that allow for thorough monitoring of soil health [11], as well as temperature, moisture levels, and weather forecasts via integrated APIs. The integration of IoT with machine learning and deep learning techniques. They can practice streamlining irrigation and fertilization practices based on real-time data [12].

Raspberry Pi 5 is a critical component in this solution, collecting location-specific weather data through APIs and connected sensors that collect real-time soil nutrient parameters. The central processing unit of the Raspberry Pi 5 processes data using advanced machine learning and deep learning models [13] to generate crop recommendations, which are displayed on an intuitive dashboard. This IoT-enabled architecture ensures real-time responsiveness and practical usability for farmers by providing on-field recommendations from the Pi 5.

The Internet of Things (IoT) extends beyond mere data acquisition by enabling automation and control mechanisms for farmers. In this work, for instance, the delivery of real-time soil metrics empowers the Raspberry Pi controller to regulate irrigation and fertilization processes. Integrating automated management of these inputs within the system directly tackles a significant limitation of conventional agriculture—namely, the excessive and inefficient utilization of resources [14]. The Raspberry Pi 5 acts as the central processing unit with ML and DL models, interfacing with soil sensors to collect data from which it can predict crops. It can also dynamically adjust irrigation and fertilizer levels by attaching actuators based on real-time field data.

This research implements a crop prediction system using machine learning and deep learning techniques. We also built an IoT system to provide real-time crop predictions. The significant contributions of this work are as follows:

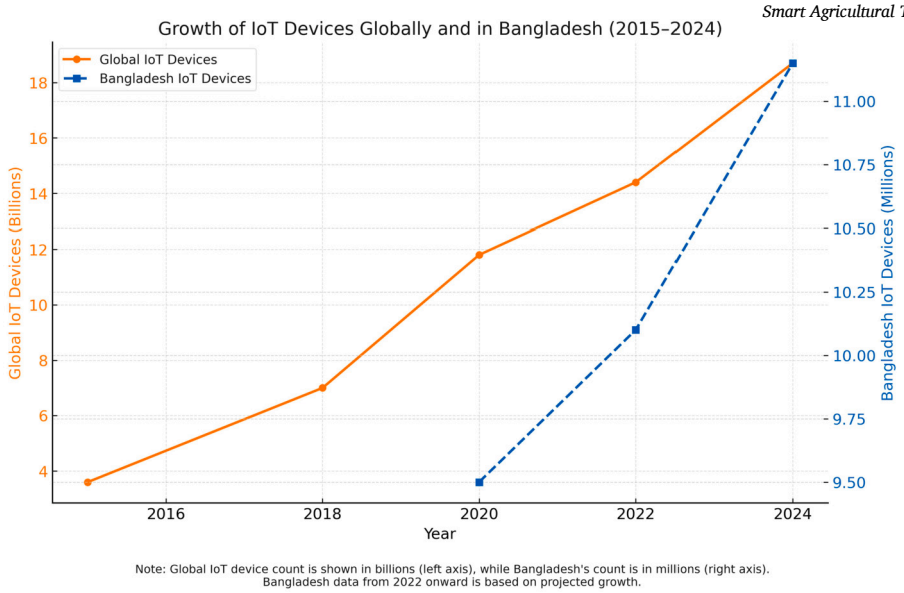


Fig. 1. Global and Bangladeshi growth of IoT over time.



Fig. 2. Major cyclonic storms in Bangladesh (2016–2024), showing affected regions and wind speeds, highlighting coastal vulnerability to climate change.

- The private dataset employed in this work contains 3,300 samples of 22 crops with eight soil nutrition and environmental features obtained from the Soil Research Development Institute (SRDI) and Bangladesh Agri-Meteorological Information Service (BAMIS), which were collected from eight unions of Habiganj district.
- A state-of-the-art utilization of stacking by combining the predictive strengths of multiple machine learning classifiers, we were able to get models that performed better at predicting crops than individual models.
- Integration with IoT for Real-Time Crop Prediction: Our best-performing model was deployed into our IoT system, primarily consisting of Raspberry Pi 5 and other sensors, enabling farmers to make real-time predictions using our model.

To the best of our knowledge, this is the first time a study on real-time IoT-based crop prediction using machine learning and deep learning has been done using a dataset of 22 crops from Bangladesh.

The rest of the article is organized as follows: Section 2 provides a critical review of recent crop prediction research, highlighting existing gaps and the rationale for our approach. Section 3 outlines the methodology, including data collection, preprocessing, and feature engineering. Finally, Section 4 offers a comparative analysis of the AI models and details the design and implementation of the IoT prototype.

2. Literature review

Usually, crop selection for yielding is chiefly grounded in farmers' experience [15], traditional and intuitive local methods, causing ineffective resource allocation and a decline in overall productivity [16]. Integrating IoT [17] with sophisticated and advanced artificial intelligence techniques [18] is gaining significant recognition in precision agriculture on a global scale. Comprehensive research attests to how such integration facilitates real-time soil and environmental data analysis, thereby improving decision-making [19]. Current research corroborates the revolutionary potential of IoT and AI-based technologies in precision agriculture [20], augmenting sustainable and effective farming systems. This section highlights the contributions of ML and DL to crop recommendation systems and assesses cutting-edge advancements in IoT-driven agricultural prediction technologies.

Real-time crop and fertilizer predictions were developed using an IoT system with machine learning techniques [21]. For instance, Islam et al. [22] combined a Kaggle dataset with a regionally diverse dataset collected from Bangladesh. The combined dataset underwent preprocessing using chi-squared tests and Z-scores to eliminate irrelevant features and outliers. Among the models tested, the CatBoost Classifier achieved the highest accuracy in crop recommendations at 97.5%, while Random Forest with GridSearchCV attained an excellent 99.56% accuracy for fertilizer prediction. Shams and colleagues [23] developed the XAI-CROP algorithm, which includes functionality for explaining why crops are recommended. After integrating LIME with XAI-CROP, their recommendation system achieved an explainability score of 94.15%. This offers reliable and easy-to-understand crop suggestions. Hasan et al. [24] proposed a machine learning model to tackle the country's agricultural dependence. The study used a dataset of over 7,000 records covering

Table 1
Annual Crop Losses Due to Floods and Cyclones.

Year	Total Flood Loss (MT)	Total Cyclone Loss (MT)	Overall Loss (MT)
2017–18	913,813	0	913,813
2019–20	0	195,079	195,079
2020–21	56,293	1,002	57,295
2021–22	76,752	8,678	85,430

Table 2
Annual Weather Statistics of Bangladesh (2013–2023).

Year	Mean Temp. (°C)	Max Temp. (°C)	Rainfall (mm, Nat'l Avg.)	Relative Humidity (%)
2013	25.6	36.7	2,427	78.0
2014	25.4	36.2	2,074	77.6
2015	25.8	37.0	2,796	79.2
2016	26.2	37.2	2,187	78.9
2017	26.5	37.7	2,597	80.3
2018	25.9	36.9	2,724	77.7
2019	25.7	36.6	2,648	77.5
2020	25.8	36.8	2,733	78.8
2021	26.0	37.1	2,921	80.1
2022	26.1	37.3	2,478	77.9
2023	26.4	37.5	2,610	78.7

five major crops: Aus rice, Aman rice, Boro rice, potato, and wheat. They analyzed about 1,400 samples per crop collected from various districts across Bangladesh. The results showed excellent accuracy; the obtained crop predictions had MSE and R^2 values as follows: Aus: 0.009 MSE, 99% R^2; Aman: 0.92 MSE, 90% R^2; Boro: 0.246 MSE, 99% R^2; wheat: 0.062 MSE, 99% R^2; potato: 0.016 MSE, 99% R^2. Priyanka and her team [25] created an IoT-enabled crop recommendation system using machine learning techniques for smart agriculture. They compiled their dataset from various sources, including government databases, Kaggle, and Google's weather forecasting services. Data preprocessing included a correlation matrix. Employing Random Forest and Decision Tree classifiers, the system achieved crop recommendation accuracies exceeding 95%, with both models reaching up to 99.77%. Ultimately, they created a smartphone app using the Kodular Creator platform to provide crop recommendations.

Bilal et al. [26] introduced a dual-phase ML-based framework to optimize crop production—the first phase comprised data gathering about farming and generating productivity estimates. The second phase developed a transfer learning model for weeds, herbicides, and crop diseases. This enables farmers to act almost immediately on early warnings with mean accuracies of up to 98%, boosting overall production outcomes.

Raju et al. [2] improved crop prediction performance by combining stacking ensemble techniques with machine learning. This effectively handled the weaknesses of traditional statistical methods. The ensemble methods boosted model accuracy by lowering variance and bias. They also reduced overfitting through independent data distribution and improved generalization. Their proposed IML-ASE framework is a multi-layered machine learning and stack ensemble model. It combined several base classifiers, including SVM, Decision Tree, Random Forest, XGBoost, AdaBoost, and Naive Bayes. A second-layer meta-learner fine-tuned the combined predictions, while a third-layer fine-tuner created the final outputs. The IML-ASE model reached an F1 score of 97.09% for multiclass crop prediction.

Musanase et al. [27] developed an integrated crop and fertilizer recommendation system to optimize agricultural practices for smallholder farmers in Rwanda. The system delivered precise, context-aware recommendations by combining IoT technologies with artificial intelligence. Data were sourced from multiple origins, including Kaggle, the Rwandan Soil Nutrient Balance Dataset, scientific literature, and expert input. Their Crop and Fertilizer Recommendation System (CFRS) achieved an accuracy of 97% using a dataset comprising 10,440 samples.

Mahale et al. [28] created a crop recommendation and weather forecasting system for Maharashtra, India, using machine learning and deep learning classifiers. They obtained the dataset from the India Meteorological Department, which includes data from 2001 to 2023. The crop dataset from Kaggle included 22 classes. A GUI-based system recommended the five best-fit crops per division in Maharashtra. The Random Forest classifier yielded the best results, with an accuracy of approximately 92%. Manju and colleagues [29] developed an IoT-based crop prediction system using machine learning classifiers. They looked at soil features like potassium, phosphorus, nitrogen content, temperature, pH, moisture content, and electrical conductivity to classify four crops: coconut, ginger, plantain, and tapioca. Of the methods used, including SVM, k-nearest neighbors, and artificial neural networks, the KNN classifier performed the best, achieving an accuracy of 84

Neha et al. [30] developed a crop predictive system based on the Arduino hardware kit. Using a Kaggle dataset comprising eight features and 22 crop types, the Random Forest algorithm achieved an accuracy of 99.54%. While collecting real-time environmental data with sensors, a Flask-based webpage provided nutrient management suggestions tailored to specific crops. Bouni et al. [31] initiated an IoT-based machine learning system for crop recommendation and yield prediction. They preprocessed data with normalization and exploratory analysis on a dataset exceeding one million data points, including temperature, humidity, soil moisture, and nutrient levels. An ensemble model combining LightGBM, Decision Tree, and Random Forest achieved the best accuracy of 99.31%. Sensors provided real-time data to Firebase, and a mobile app was developed to advise farmers on crop selection and resource management.

Prity et al. [32] created a crop recommendation system based on machine learning to improve agricultural yield. They used Kaggle datasets that included soil nutrient data, pH, rainfall, temperature, and humidity. The Random Forest technique achieved the highest accuracy at 99.31%, with Extra Trees and Gradient Boosting following closely behind. The authors incorporated XAI techniques such as LIME, SHAP, Integrated Gradients, and Layer-wise Relevance Propagation to enhance interpretability, evaluating parameters such as N, P, K, weather, and soil pH about recommendations. Akkem et al. [33] demonstrated via a Streamlit-based dashboard how incorporating Explainable Artificial Intelligence (XAI) mechanisms in crop recommendation systems improved farmers' trust by addressing the so-called 'black box' problem. Counterfactual explanations generated using 'dice-ml' allowed farmers to ask 'what if' questions and make decisions based on better outcomes. The paper evidences the prospects of XAI, showing how it enhances Decision Tree algorithms and ensemble models, including CNN deep learning for monitoring crop health.

Table 3
Summary of the Employed Private Crop Prediction Dataset.

	N	P	K	pH	Temp (^0C)	Humidity (%)	Moisture (%)	Rainfall (cm)
count	3300	3300	3300	3300	3300	3300	3300	3300
mean	223.728	57.8658	81.8218	5.85464	25.86	70.8905	65.5912	100.575
std	70.6896	23.4862	66.3779	0.734418	5.73982	6.91895	20	40.1172
min	55	14	9	3.3	13	55	12	40
25%	160	43	39	5.6	22	66.2	60	67.75
50%	210	49	61	6.1	26	71.3	70	95
75%	275	78	93	6.3	30	75.6	79	127
max	399	444	450	6.7	40	85.9	95	250

Sharma and Shivandu [11] introduced an AI-enabled IoT architecture for precision agriculture that couples remote-sensing imagery, differential GPS (DGPS), and autonomous AgroBots to curb over-harvesting and automate weed detection. A major challenge is scaling various datasets and tailoring solutions to fit smallholder farm conditions. This remains crucial for creating strong AI models that can operate under multiple environmental conditions, an ongoing issue in the overall effort for sustainable and efficient precision agriculture.

Similarly, Wang and colleagues [34] developed an IoT-based system to reduce crop diseases and insect infestations using ozone sterilization and light-trap technology. Their system featured real-time environmental sensors and a mobile app for remote management. In controlled trials, the system cut the incidence of powdery mildew in cucumbers by 36.68% compared to untreated crops. This shows its effectiveness in both greenhouse and open-field settings.

Edge-AI deployments in field agronomy are converging across complementary applications, offering promising pathways toward a more sustainable and resilient agricultural future. Ultra-light onboard inference—such as YOLOv5s compressed into a 9.7 MB MobileNet–GhostNet variant (YOLOv5s-Im)—runs at 22 FPS on a Jetson Xavier NX, guiding a quadcopter pollination rig and quadrupling successful blossom contacts versus heavier baselines. Fastellini et al. [35] showed that cloud-linked sensor swarms, GNSS-guided machinery, and data-fusion dashboards already enhance input-use efficiency and yield stability in vineyards, arable crops, and smart irrigation pilots. This provides a blueprint for the ThingsBoard-centered architecture proposed here, which uses energy-autonomous nodes. Portilla et al. [5] described roll-to-roll printed TFT tags that scavenge RF and indoor-light energy to sustain μW-class sensors, foreshadowing battery-less upgrades to the Raspberry Pi–based edge system proposed here.

Collectively, these works ground the real-time, explainable crop recommendation platform in the state of the art in lightweight AI models, scalable IoT infrastructure, and self-powered hardware.

So far, crop recommendation studies have mainly relied on generalized public datasets with limited regional focus regarding samples, features, or crop diversity, which reduces overall generalizability and scalability. Additionally, they often neglect the unique environmental conditions, agroecological zones, and agricultural challenges that regions such as Bangladesh face—focusing on crop recommendations without incorporating sensor and weather data results in a lack of real-time integration and on-field crop suggestions. Few systems developed in the past year have emphasized explainability in AI models to build trust among agricultural practitioners.

Driven by these deficiencies, this work bridges the gap by focusing on a Bangladesh-specific dataset and integrating real-time soil and weather data using sensors and weather APIs. Advanced deep learning architectures—such as TabNet, CNN-BiLSTM—and state-of-the-art ensemble machine learning techniques demonstrate high prediction accuracy. Greater transparency is achieved through explainable AI techniques; for instance, LIME enhances the interpretability of model predictions. The developed IoT prototype demonstrates feasibility through real-time deployment, emphasizing practical utility for precision agriculture. On the ThingsBoard platform, an intuitive interface facilitates seamless access to real-time monitoring data and crop recommendations, promoting accessibility and usability.

3. Methodology

3.1. Dataset

The private dataset used in this research was collected from multiple surveys and analyses, published by the Soil Resource Development Institute (SRDI) [36], and contains crop data for eight unions in Habiganj District, Sylhet, Bangladesh. These data were meticulously collected in the SRDI lab from multiple soil samples in that region. Weather data related to crops, such as temperature, rainfall, and humidity, were collected from the Bangladesh Agro-Meteorological Information Service (BAMIS), which supplies crop-wise meteorological data. The dataset contains the soil properties of pH, nitrogen, potassium, and phosphorus content of 22 crops. The dataset is quite balanced with 150 samples for each crop, eight features, and a total of 3300 regional crop samples.

Table 3 presents a detailed statistical analysis of the numerical, non-discrete features in the dataset, including the count, mean, standard deviation, and other relevant descriptive statistics. The wide range of values observed for most features, such as rainfall (40 cm to 250 cm) and temperatures (13 to 40 degrees Celsius), suggests a highly varied distribution within the dataset.

NPK nutrient analysis in Fig. 3 indicates different patterns of distribution among different crop types in Bangladesh. The box plots show uniform nitrogen needs between 150 kg/ha and 300 kg/ha for almost all crops. On the other hand, it was observed that phosphorus has lower overall needs (20-80 kg/ha) but higher inter-crop variability, whereas potassium shows the highest variation (20-200 kg/ha) with wide differences among crop categories. These patterns of nutrient distribution justify the use of NPK as salient features in our machine learning model. The high variability, especially of potassium concentration, offers strong discriminatory power for the classification of crops. Outliers in phosphorus indicate crop-specific nutritional needs that can be used as unique markers for precise crop prediction.

Fig. 4 shows the kernel density estimation (KDE) plots and probability distributions of NPK throughout the dataset. Phosphorus comprises a sharp, narrow peak at 40-50 kg/ha, indicating consistent demands with minimal variation. It is observed that Potassium has a wider distribution at 50-80 kg/ha, whereas nitrogen has the widest distribution with several peaks up to 400 kg/ha. The multi-modal distribution indicates varied crop nutritional demands, confirming NPK nutrients as good discriminative features for the classification of crops.

Fig. 5 demonstrates the color-coded bubble plots by crop type, showing the diversity of growing conditions across crops. These plots illustrate the multimodal nature of environmental preferences.

Fig. 6 illustrates the class distribution of various crops in the dataset. As shown, each crop has an equal number of instances, resulting in a well-balanced dataset.

3.2. Data preprocessing

To preprocess the dataset, duplicate instances and null values were addressed. Min-Max feature scaling was applied to standardize numeri-

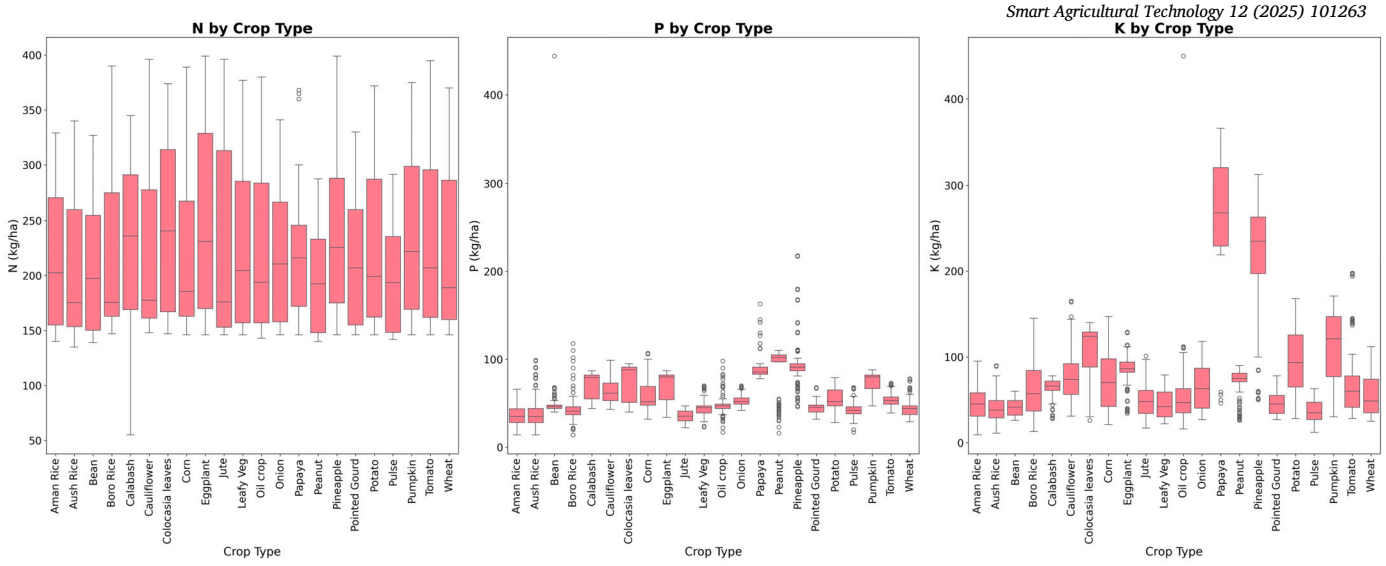


Fig. 3. NPK boxplots of the employed crop prediction dataset.

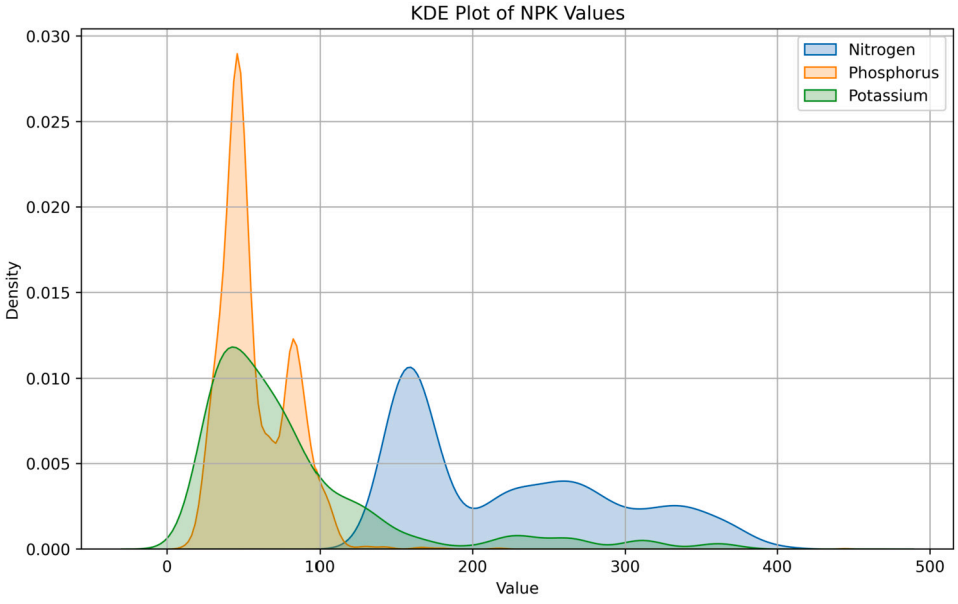


Fig. 4. NPK KDE plot representation of the employed crop prediction dataset.

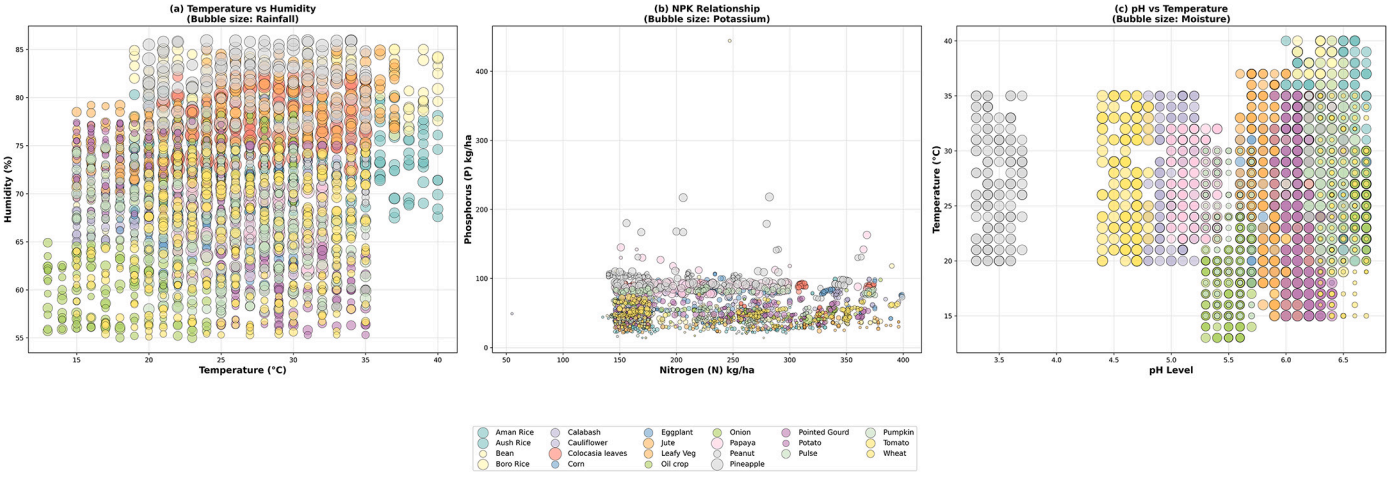


Fig. 5. Bubble plots for environmental features of the employed crop prediction dataset.

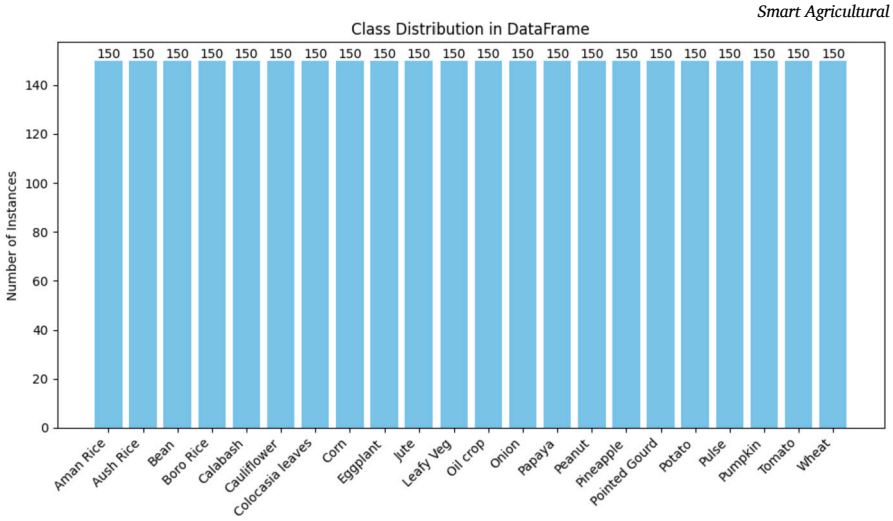


Fig. 6. Class distribution of crops.

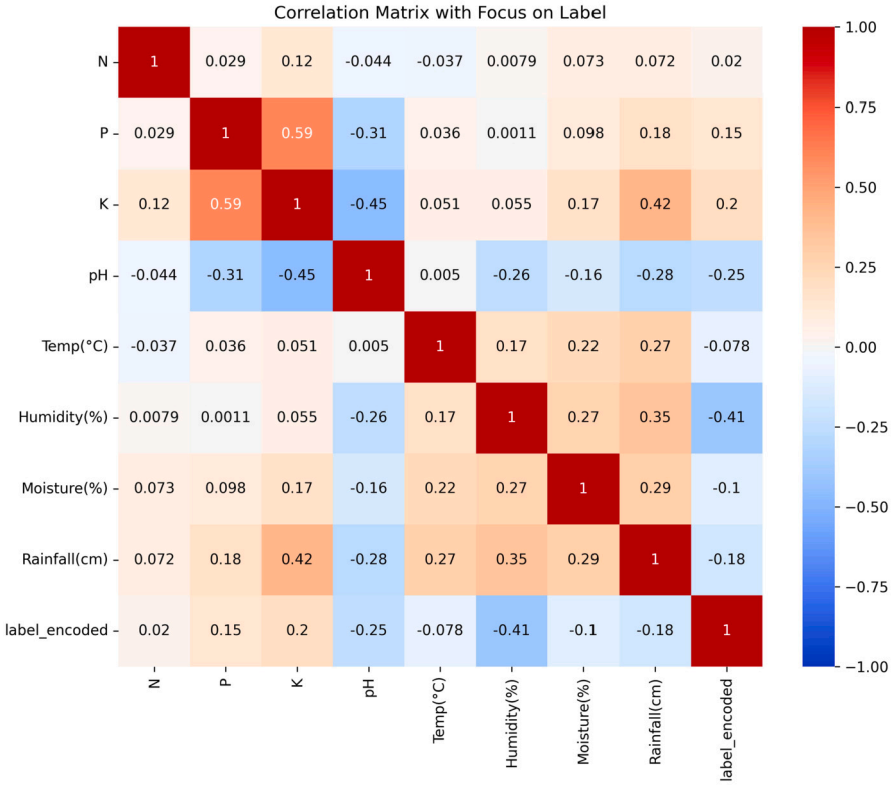


Fig. 7. Correlation heat map of various features of the employed crop prediction dataset.

cal features within the range [0, 1], ensuring equal contribution of each feature during model training. Label encoding was employed to convert categorical labels into numerical representations, thereby ensuring compatibility with machine learning algorithms.

Fig. 7 displays a heatmap of feature correlations of the employed crop prediction dataset. The highest correlation (0.59) was observed between Potassium (K) and Phosphorus (P). All features were retained for analysis as this value is below the established threshold of 0.80, indicating multicollinearity.

The correlation heatmap (Fig. 7) reveals the strongest correlation between potassium and phosphorus, while pH and temperature exhibit the weakest. These variations in correlation are visually represented by the kernel density estimate plots in Fig. 8, where the high degree of overlap between potassium and phosphorus contrasts sharply with the minimal overlap between temperature and pH. Next, the dataset was partitioned in an 80:20 ratio, with 80% allocated for training and the remaining 20% reserved for testing.

3.3. Machine learning classifiers used

- A Decision Tree classifier is a supervised learning technique used to classify and analyze regression problems. It operates by recursively partitioning the feature space into partitions depending on the input feature value and building homogeneous partitions about the target variable. The Decision Tree nodes correspond to the feature attributes, branches represent decision rules, and leaf nodes

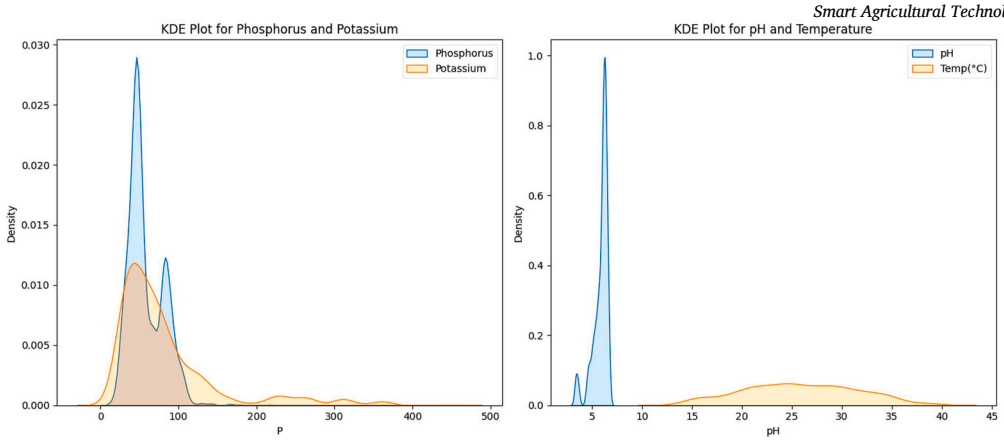


Fig. 8. KDE plots comparing nutrient and environmental variables.

correspond to the labels that must be predicted. When creating the decision tree, the goal is to determine where to split the data by feature at the nodes, often based on information gain, Gini impurity, or entropy.

- Random Forest Classifier works on the ensemble learning method. It works by designing several Decision Trees during training time. These trees are trained on the subset of the training data and input features. The output from all the trees is aggregated to get the final output.
- The KNN algorithm is a well-known supervised learning classifier that uses the neighborhood and surrounding data points to make predictions about the grouping of individual data points. It works on the principle that similar data points are found near one another in the same feature space. The variable k denotes the number of neighbors considered during the prediction of new data points. In this study, hyperparameter tuning resulted in an optimal k value of 3, achieving an accuracy of 97.5%.
- SVC is an extraordinary machine learning algorithm that creates hyperplanes to separate data classes. It can handle high-dimensional data and is handy when data can not be separated using straight lines.
- Adaptive Boosting (AdaBoost) is an ensemble learning framework that combines many weak learners to create strong learners for classification problems.
- Gradient Boosting is used for regression and classification tasks and produces an ensemble of shallow trees that sequentially correct its preceding model's mistakes. It works on an initial prediction, calculates residuals, fits new trees onto them, and updates the predictions iteratively. Instead, the key aspects include the loss function, base learners (usually shallow trees), learning rate, the number of trees, and regularization techniques such as tree constraints, shrinkage, and subsampling to avoid overfitting. Examples of well-tuned implementations are XGBoost, LightGBM, and CatBoost. Gradient Boosting is extremely fast and flexible but can also quickly overfit the model and requires care in tuning the hyperparameters.
- Categorical Boosting (CatBoost) is an advanced gradient-boosting algorithm that is highly effective at handling categorical data, which makes it especially good at applications such as recommendation systems, fraud detection, and customer analytics. It natively handles categorical features with very little preprocessing, so there is no need for one-hot encoding. It also uses techniques like ordered boosting to deal with overfitting while maintaining high accuracy and excellent performance results. Highly configurable, easy to use, and yet powerful in machine learning tasks, it allows model parameters to be adjusted to obtain the best results in various applications.
- Logistic Regression is a classification method that can be used to model the probability of the target variable and the function of the explanatory variables (predictor variables). This interprets linear combinations from inputs to probabilities, and utilizes a sigmoid function from the estimated logged odds to probabilities. Regularization techniques like L1 (Lasso) and L2 (Ridge) remove the overfitting problem by penalizing significant coefficients. This model function of Logistic Regression establishes the relationship of a dependent binary variable with independent variables (predictors), which in this space is logistic, that is, probabilistic space. The benefits of this approach are that it is simple, interpretable, efficient, and appropriate for prediction problems with probabilistically structured outputs and is widely used in medicine, marketing, finance, and social sciences. It has its strengths but assumes a linear relationship and is sensitive to outliers.
- The Stacking classifier is an ensemble method. It works by combining multiple base learners and training them on the same dataset. The outputs are used as input features for the meta-learner (higher-level model) classification, where the final prediction is made. Working steps of the proposed stacking classifier for crop prediction is illustrated in Algorithm 1.
- Hyperparameter tuning was performed on each model to optimize accuracy in this work. Model-specific hyperparameter sets were explored. The kernel type, gamma, and C parameters were tuned for the Support Vector Machine model. Additionally, for Logistic Regression, the penalty and solver parameters were also considered. The Decision Tree model's hyperparameters included maximum depth, minimum samples split, and minimum samples leaf. For the k-nearest neighbors model, the number of neighbors, p, and weights were tuned.

3.4. Deep learning models used

- TabNet is one of the most robust deep learning architectures specifically designed for tabular data. It employs sequential attention to select and process features at each decision, which allows the model to learn interpretable and sparse feature representations, which is why it can capture complex patterns and dependencies in the data. The proposed TabNet architecture was configured with specific hyperparameters, including decision prediction and attention embedding widths, number of decision steps, and feature reuse coefficient. Additionally, λ_{sparse} was specified for feature selection loss, and the Adam optimizer was employed with learning rate scheduling using StepLR.
- Convolutional Neural Networks (CNNs) are the most used deep learning architectures for analyzing visual images. This architecture is designed to learn automatically and update the spatial hierarchies of features through backpropagation across several layers, including convolutional, pooling, and fully connected layers. Our CNN architecture was built using convolutional layers, ReLU activation functions, and max pooling operations. The tabular data was then

Algorithm 1 Proposed Stacking Classifier for Crop Prediction.

Input: Dataset (X, y)
Output: Best Stacking Classifier Model
Step 1: Data Preparation
Split dataset into training and testing sets: $(X_{train}, X_{test}, y_{train}, y_{test})$
Step 2: Initialize Base Learners
Define and test different combinations of machine learning models as base learners:
- Random Forest (rf_tuned)
- XGBoost ($xgboost_tuned$)
- Gradient Boosting ($gboost_tuned$)
- Decision Tree (dt_tuned)
- SVC (svc_tuned)

Step 3: Define Meta-Learner
Test different models as meta-learners to find the optimal combination, including:
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- SVC

Step 4: Build Stacking Classifier
Create a stacking classifier using the best combination of base learners and meta-learner.
Step 5: Hyperparameter Optimization
Define parameter grid for tuning the meta-learner.
Perform RandomizedSearchCV for various meta-learners.
Step 6: Train and Select Best Model
Fit the model using RandomizedSearchCV.
Extract best parameters and best model configuration.
Step 7: Model Evaluation
Predict on training data and compute accuracy.
Step 8: Output Best Model Performance
Print best parameters and cross-validation accuracy.

reshaped into a time grid structure to make it compatible with the CNN's inputs. Afterward, it was converted into tensors and reshaped to retain one color channel (grayscale) while maintaining the exact dimensions. Hyperparameter optimization was done using Optuna to develop the best Crop CNN model. Leveraging the Optuna framework, an optimal set of architectural and training hyperparameters was systematically identified, encompassing the number of convolutional layers, initial channel count, kernel size, learning rate, and training epochs. The most useful configuration, as determined by the optimization process, consisted of two convolutional layers with 32 filters and a kernel size of 2. Subsequently, the feature maps were processed and flattened before being passed to a fully connected layer for final classification.

- Long Short-Term Memory (LSTM) is a type of recurrent neural network used for sequence tasks and is excellent at using long-term dependencies in data. It is particularly well-suited for tasks involving sequential data, such as time series analysis, natural language processing, and speech recognition. To enable the application of LSTM on tabular data, the dataset was reshaped by introducing a sequence dimension, ensuring compatibility with LSTM input requirements. Utilizing Optuna, the optimal model configuration was determined, including the number of hidden layers, LSTM layers, dropout rate, learning rate, and number of training epochs. Hyperparameter Optimizing helps by not making the model unnecessarily complex; our LSTM model consisted of two LSTM layers with 96 hidden units each.
- CNN-LSTM is a common hybrid deep learning neural network architecture that combines CNN and LSTM. It is suited for problems involving temporal and spatial dependencies. The data is reshaped for use with the CNN-LSTM model. The convolutional layers learn spatial features from the data, and the last layers learn temporal dependencies. Through the utilization of Optuna, a CNN-LSTM model was constructed, comprising four convolutional layers followed by a single LSTM layer with a hidden size of 64. This architecture enables sequential analysis of the feature representations produced by the flattened CNN outputs.
- The CNN-BiLSTM is a combination of Convolutional Neural Network (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) layers for classification tasks with time-series data. The network architecture commences with a one-dimensional convolutional layer that operates over length-3 segments, followed by ReLU activation and max-pooling. Those features are further reshaped and then fed through a BiLSTM that is primarily used to capture temporal dependencies on a given sequence, both in the future and in the past. The last output of the BiLSTM is passed to a dense layer for classification. Using CNNs for local feature extraction and BiLSTMs for temporal context understanding, these two-fold temporal and local attentions can effectively handle complex sequence-based classifications. Our best Optuna-optimized CNN-BiLSTM model consists of three convolutional layers with 16 initial filters and a kernel size of 2, used to extract spatial features. These features are then passed to a single-layer bidirectional LSTM with 99 hidden units for sequential learning.
- Local Interpretable Model-Agnostic Explanations (LIME) is the most powerful explainable artificial intelligence (XAI) technique. It provides interpretability for black-box machine learning models and serves as a visualization technique that helps explain individual predictions. This approach is advantageous because it treats any supervised learning model as a black-box." In the present implementation, a pipeline was constructed comprising principal component analysis (PCA) for dimensionality reduction, followed by a RandomForestClassifier for prediction. Subsequently, a LimeTabularExplainer was instantiated with the training data, feature names, and class labels, with the mode parameter set to "classification" to accommodate the multiclass setting. The contribution of each feature to the model's prediction for the selected instance was then visualized within a Jupyter notebook.

For all deep learning models, the Optuna framework [37] was employed for hyperparameter tuning to identify the optimal architecture and number of training epochs, thereby maximizing predictive performance.

Fig. 9 illustrates the workflow of the proposed crop prediction study. The process begins with data preprocessing, including identification and handling of null and duplicate values, followed by normalization using min-max scaling. The dataset is then partitioned into 80% training and 20% testing sets for machine learning model development. Multiple machine learning algorithms undergo hyperparameter tuning, training, and testing until satisfactory performance is achieved. Deep learning methodologies, including custom models such as TabNet, LSTM, and CNN-LSTM, are applied to the dataset. The Optuna framework guides hyperparameter tuning for these deep learning models to optimize architecture and performance. Finally, the highest-performing model, selected based on inference time, is deployed to the IoT device for real-time crop prediction.

3.5. Hardware interfaces

The proposed solution integrates the best-performing AI model with a Raspberry Pi 5 for real-time crop recommendations. As depicted in Fig. 10, a 7-in-1 soil sensor collects real-time soil data (N, P, K, pH, temperature, and moisture) via an RS485 interface. This data is transmitted to the Raspberry Pi 5 via a USB to RS485 converter. The pre-trained AI model (stored as a .pkl file) on the Raspberry Pi 5 processes this soil data along with real-time weather parameters (humidity and rainfall) retrieved from a weather API. The model then predicts the most sustainable crop for the given soil and weather conditions, and the results are displayed on the ThingsBoard IoT platform.

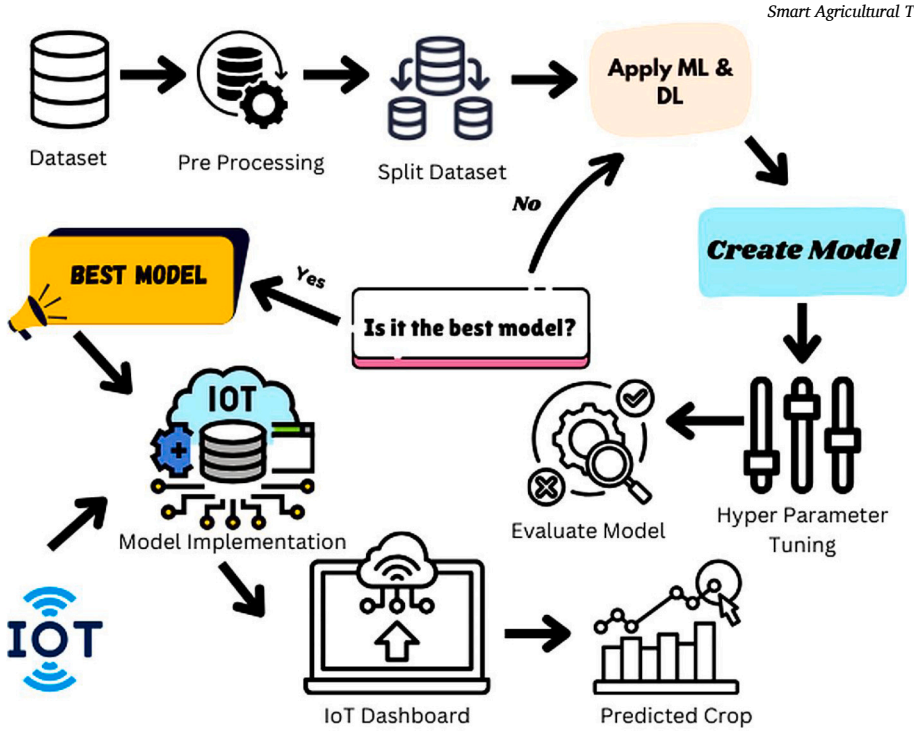


Fig. 9. Working sequences of the proposed crop prediction system.



Fig. 10. Hardware Setup of the proposed crop prediction device.

- The Raspberry Pi 5 is a high-performance computing platform that is ideal for processing any AI model with optimization. It has dual-band 802.11ac Wi-Fi, making it the best choice over NodeMCU [38] and Arduino. The microSD card slot supports the high-speed SDR104 mode and can store any machine and deep learning model.
- 7 in 1 Agriculture Sensor is a multiparameter five-pin RS-485 interfaced soil nutrient (N, P, K, pH, temperature, moisture) measuring sensor that is highly precise, corrosion resistant, and IP68 rated. It is ideally used for scientific research, irrigation systems, and precision agriculture.
- USB to RS485 converter provides seamless data transmission between the USB and RS485 interfaces, making it reliable for transferring data from any half-duplex sensor to any USB device.

This paragraph discusses the functions and costs of various components for the proposed prototype design. The Raspberry Pi 5 is the central processing unit for predictions and the sensor data storage device, which costs approximately 14,500 BDT ($118). Including a compact casing and cooler, the total cost of the Raspberry Pi 5 amounts to 20,000 BDT ($163). The sensor, along with a Modbus converter and other components, costs approximately 16,500 BDT ($134). Thus, the total cost of the entire setup is 36,500 BDT ($297). Furthermore, a power bank with a capacity of 20,000 mAh is included at an additional cost of 1,500 BDT ($12), allowing the Raspberry Pi 5 to operate for approximately 8 to 9 hours in practice.

Fig. 11 shows the data from the 7-in-1 sensor, and the readings from the holding registers were retrieved using the Modbus-RTU protocol while converting the RS485 interface to USB with our converter for integration with the Raspberry Pi. The Python pymodbus library was installed to scale and process raw sensor readings in the device, then converted moisture values into percentages and temperature readings into floating-point numbers. The processed data were stored in a .csv file

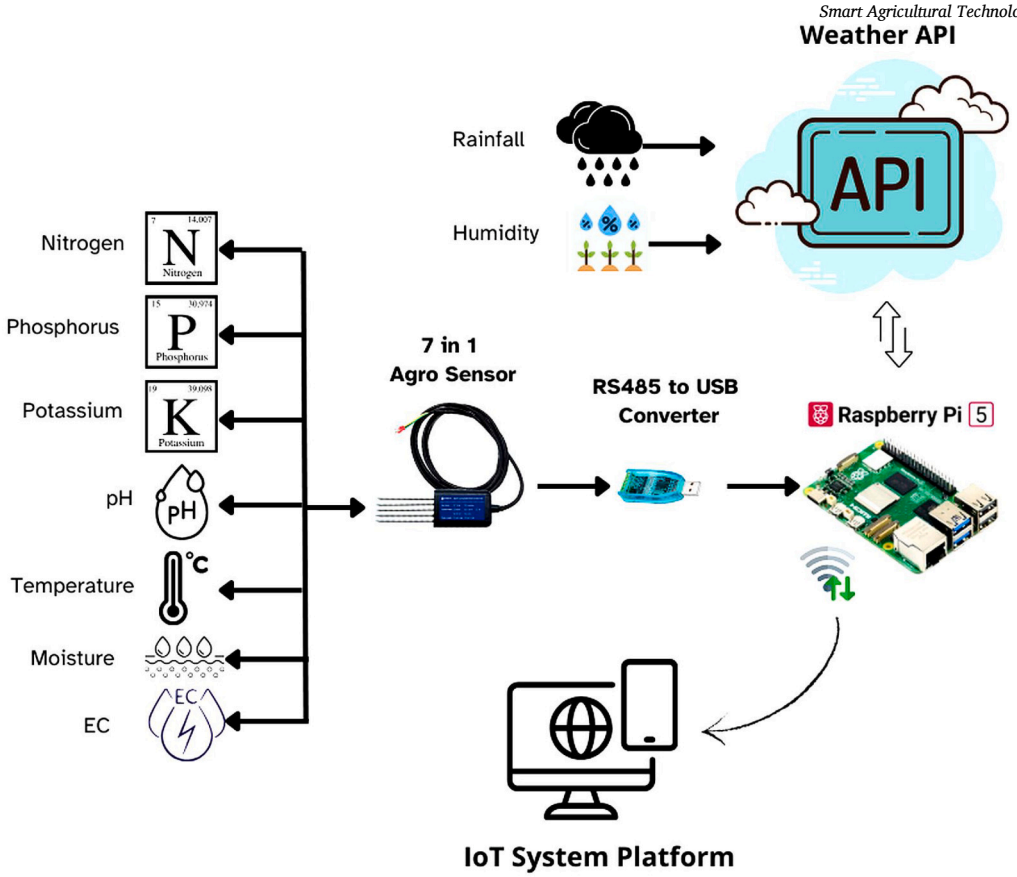


Fig. 11. IoT Operational Workflow of the developed system.

along with corresponding timestamps for offline storage and analysis. A one-minute delay was added to the system to allow for sensor stabilization and to improve processing. Real-time weather parameters, such as humidity and rainfall, were retrieved through HTTP requests using location data from the WeatherAPI service. This dynamically fetches current humidity levels and average monthly rainfall values. The Python joblib library was used to load a pre-trained scaler, saved as a .pkl file, to standardize both sensor and weather data. This ensured consistency with the data used during model training. A label encoder was also loaded to change categorical labels into numeric form.

The best-performing pre-trained model `random_search_model.pkl` was successfully deployed on the Raspberry Pi for real-time crop prediction. The model received scaled soil and weather data as input and returned the most suitable crop based on the given conditions. The predicted crop, along with the scaled soil data, was transmitted from the Raspberry Pi to the ThingsBoard IoT platform [39] using HTTP POST requests in JSON format, with timestamps attached to ensure accurate time tracking.

For the IoT system platform (Fig. 12), ThingsBoard was selected due to its user-friendly dashboard, extensive widget customization options, and ability to display string-type outputs, such as crop names, unlike some platforms limited to numeric values. Its seamless integration with Raspberry Pi and support for device grouping were also key factors. ThingsBoard's remote monitoring and control capabilities provide global access. Furthermore, its alarm system allows for configuring alerts based on critical agricultural parameters, such as extreme pH levels, high soil temperature, or low moisture content. It enables timely responses to anomalies or threshold violations and facilitates the maintenance of optimal conditions.

The proposed AI and edge device integrated automatic crop prediction system was developed under several design assumptions: (1) low-bandwidth rural settings requiring offline inference; (2) varied soil and environmental conditions needing robust modeling; (3) the importance of explainability for trust and adoption. Insights gained from model performance benchmarking and user evaluations informed multiple refinements, including: (a) replacing high-latency models with faster ones for deployment, (b) improving model interpretability via LIME visualizations, and (c) modifying the sensor data smoothing interval for stability. These iterative changes enhanced the system's usability, responsiveness, and prediction quality.

3.6. Model evaluation

Following initial model training, each model underwent hyperparameter tuning to optimize performance. Subsequently, confusion matrices were generated for each model, and performance was evaluated based on accuracy, precision, recall, and F1-score.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \tag{1}$$

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{2}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{3}$$

4. Results and discussion

This section presents the results and analysis of the proposed machine learning and deep learning models for the proposed crop prediction system. A detailed comparison of machine learning classifier performance is provided, with and without hyperparameter tuning. The results from five different deep learning architectures—TabNet, CNN, LSTM, CNN-LSTM, and CNN-BiLSTM—optimized using the Optuna hyperparameter tuning framework are also presented. Finally, a comparison is demonstrated between the results obtained in this study and those

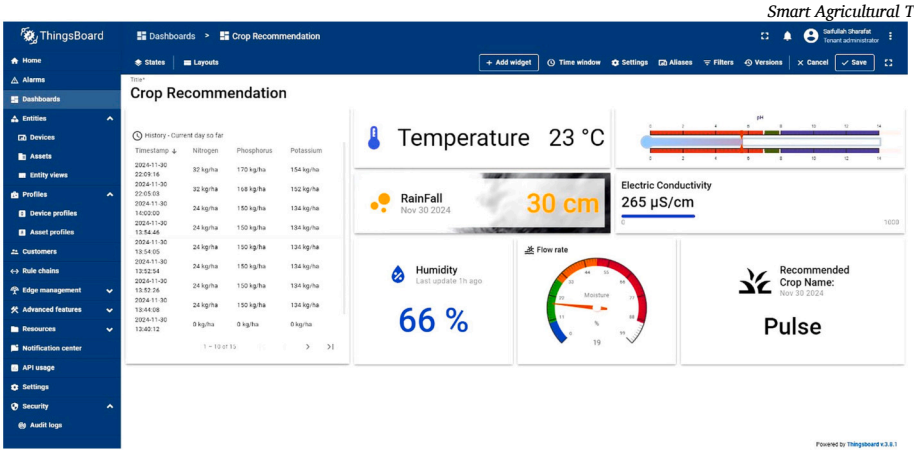


Fig. 12. Proposed real-time IoT crop recommendation system's dashboard.

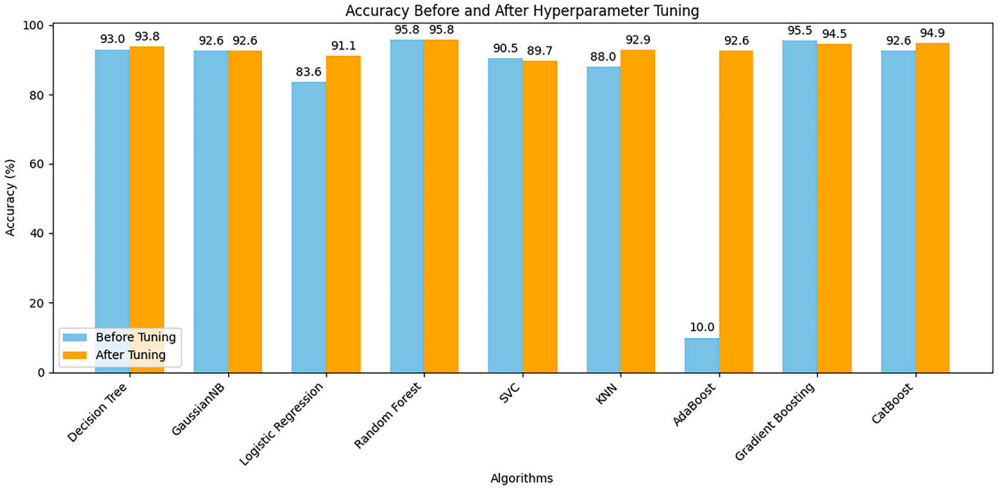


Fig. 13. Accuracies of the applied models with and without hyperparameter tuning.

Table 4
Detailed Evaluation on Test Set Using Default Classifier.

Model	Accuracy	Precision	Recall	F1-score
GaussianNB	0.926	0.930	0.926	0.926
KNN	0.880	0.894	0.880	0.882
Decision Tree	0.930	0.933	0.930	0.930
Random Forest	0.958	0.960	0.958	0.958
Logistic Regression	0.836	0.844	0.836	0.826
SVC	0.905	0.912	0.905	0.905
AdaBoost	0.100	0.070	0.100	0.049
Gradient Boosting	0.955	0.956	0.955	0.955
CatBoost	0.926	0.960	0.960	0.960

reported by other researchers in recent state-of-the-art crop prediction systems.

Table 4 summarizes the performance of eight baseline machine learning classifiers—GaussianNB, KNN, Decision Tree, Random Forest, Logistic Regression, SVC, AdaBoost, Gradient Boosting, and CatBoost—before hyperparameter tuning. Random Forest and Gradient Boosting achieved the highest performance, with accuracies of 95.8% and 95.5%, respectively. CatBoost and Decision Tree exhibited similar performance, each achieving approximately 93% accuracy. GaussianNB demonstrated consistent performance across all metrics, with an accuracy of 92.6%. AdaBoost was the least performant model before tuning, with a notably low accuracy of 10%.

Fig. 13 presents a comparative analysis of the applied classifiers' accuracies before and after hyperparameter tuning. AdaBoost demonstrated the most significant improvement, increasing by over 80% to achieve a final accuracy of 92.6%. Substantial gains were also observed for Logistic Regression (from 83.6% to 91.1%) and KNN (from 88.0% to 92.5%). Minimal improvements were noted for Decision Tree and CatBoost, while GaussianNB, Random Forest, and Gradient Boosting showed negligible changes in performance.

Table 5 presents a performance evaluation of various meta-learners after stacking with diverse base learners, detailing both accuracy and best cross-validation accuracy. The highest accuracy (95.9%) was achieved by SVC, closely followed by Random Forest (95.5%) and GaussianNB (94.5%). The cross-validation accuracy for SVC (94.8%) was consistent with its overall accuracy. While most meta-learners demonstrated stable performance across both test and validation sets, some variation was observed, such as the Decision Tree's accuracy of 93.5% on the test set compared to 88.0% during cross-validation. Overall, these results highlight the effectiveness of stacking in improving model performance, with SVC emerging as the most reliable meta-learner in this configuration.

Fig. 14 illustrates the stacking architecture of the best-performing Stacked SVC model. In this configuration, SVC serves as the meta-learner, while Random Forest, Gradient Boosting, and CatBoost function as the base learners. The figure demonstrates how the knowledge acquired by the base learners is transferred to the meta-learner to achieve improved prediction accuracy.

Fig. 15 showcases the performance of the stacking ensemble classifier with SVC as the meta-learner, achieving an accuracy of 95.9%. The 22 ROC curves, one for each class, exhibit a perfect AUC of 1.00,

Table 5
Performance Metrics of the Applied Stacking Ensemble Models.

Meta Learner	Base Learners	Test Accuracy	Best Cross Validation Accuracy
Decision Tree	RF, GB, CatBoost	93.5%	88.0%
Random Forest	DT, GB, CatBoost	95.5%	90.9%
GaussianNB	RF, GB, CatBoost	94.5%	92.5%
KNN	RF, GB, CatBoost	94.0%	92.0%
Logistic Regression	RF, GB, CatBoost	94.5%	89.9%
SVC	RF, GB, CatBoost	95.9%	94.8%
AdaBoost	RF, GB, CatBoost	90.5%	89.0%
Gradient Boosting	RF, DT, CatBoost	92.5%	89.8%
CatBoost	RF, DT, GB	93.2%	93.1%

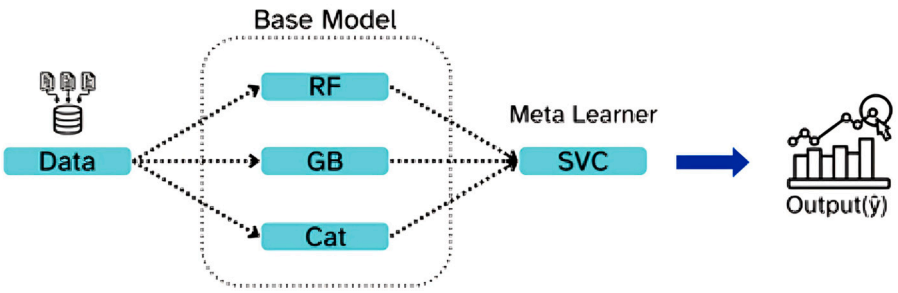


Fig. 14. Proposed SVC meta learner-based stacking architecture.

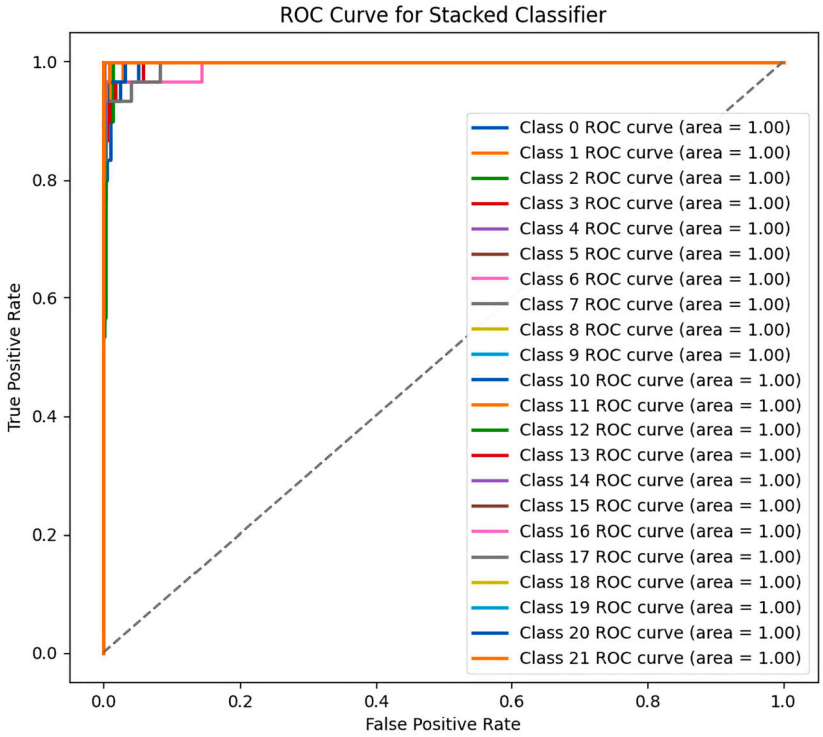


Fig. 15. ROC curve of the stacked ensemble model.

indicating excellent class separability. The curves' confinement within the upper-left quadrant reflects high true-positive rates and remarkably low false-positive rates. While these high AUC scores suggest strong classification performance, they also warrant consideration of potential overfitting.

Fig. 16 depicts the confusion matrix of the proposed SVC meta-learner-based stacking ensemble technique for automatic crop prediction across 22 different classes. According to this figure, most correct classifications fall within diagonal cells, indicating the model's strong predictive performance for certain crops. Although some misclassifications occur, they are minimal. Hence, the model demonstrates strong generalization capability on unseen data and serves as a suitable candidate for integration within the proposed IoT-based crop prediction system.

Table 6 provides a performance comparison of the deep learning architectures considered, using different metrics such as accuracy, precision, recall, F1-score, model size, number of trainable parameters, and training time per epoch. TabNet achieved the highest accuracy (92%) and proved to be the top-performing model. It also demonstrated reasonable resource requirements, with a training time of 1.21 seconds per epoch on the smallest device in the evaluation set. LSTM closely followed TabNet, achieving 91.8% accuracy. CNN reached 89.6% accu-

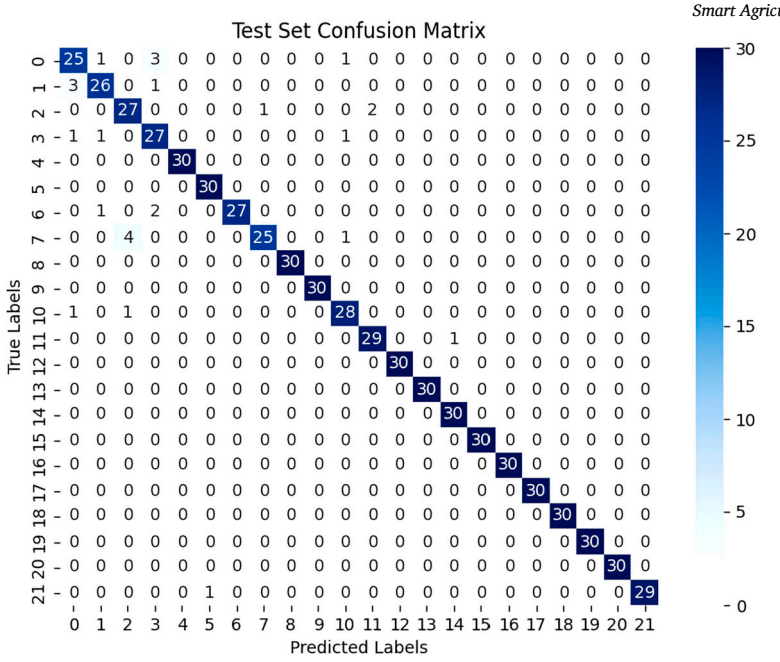


Fig. 16. Confusion matrix of the stacked ensemble model.

Table 6
Performance Metrics of the Applied Deep Learning Models.

Model	Accuracy	Precision	Recall	F1-Score	Model Size	Trainable Parameters	Avg. Training Time/Epoch
TabNet	92.0%	0.926	0.920	0.920	1.78 MB	444400	1.21 s
CNN	89.6%	0.897	0.896	0.896	0.08 MB	21206	0.38 s
LSTM	91.8%	0.926	0.918	0.920	0.45 MB	117334	0.37 s
CNN-LSTM	88.9%	0.901	0.889	0.890	0.09 MB	22582	0.41 s
CNN-BiLSTM	86.8%	0.869	0.868	0.867	2.12 MB	530050	1.54 s

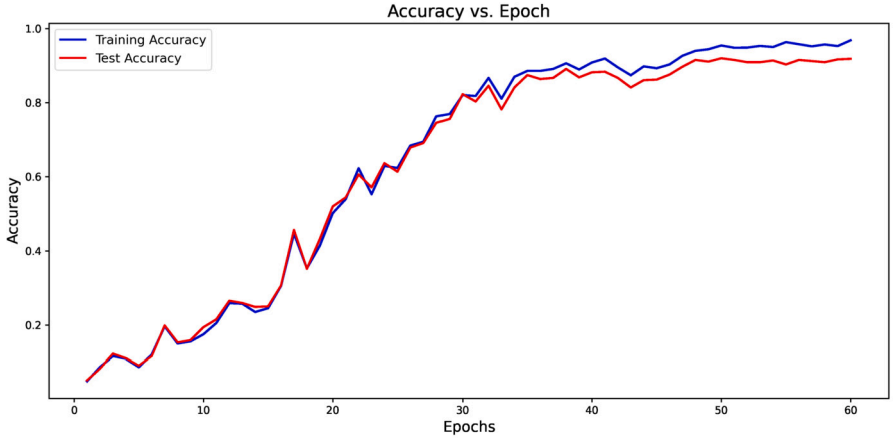


Fig. 17. Accuracy and loss vs. epochs of the applied TabNet model.

racy with a relatively small model size of 0.08 MB. The CNN-LSTM and CNN-BiLSTM models exhibited lower accuracies of 88.9% and 86.8%, respectively. The largest model, CNN-BiLSTM, had a size of 2.12 MB and comprised 530,050 trainable parameters.

Fig. 17 displays the training and testing accuracy per epoch for the TabNet model. The model trained for approximately 65 epochs before early stopping was activated. Training accuracy steadily increased, reaching approximately 95%. A similar trend was observed for testing accuracy, albeit at a lower final value of around 90%. Fluctuations in accuracy, particularly during the initial epochs, gradually diminished, indicating convergent stability with continued training. This stable performance enhances the model's reliability in consistently achieving high accuracy on both training and testing data.

Fig. 18 illustrates the training and evaluation performance of the CNN model. Both training and testing accuracy exhibited a monotonic increase, with the testing accuracy ultimately surpassing the training accuracy, reaching 92.42%. The training accuracy reached 93% around epoch 21. The loss plot demonstrates a sharp initial decrease, followed by sustained stability, indicating that the model effectively learned the underlying patterns in the data and generalized well.

Fig. 19 displays the accuracy and loss curves as a function of epochs for the custom LSTM model over 30 epochs. The training accuracy stabi-

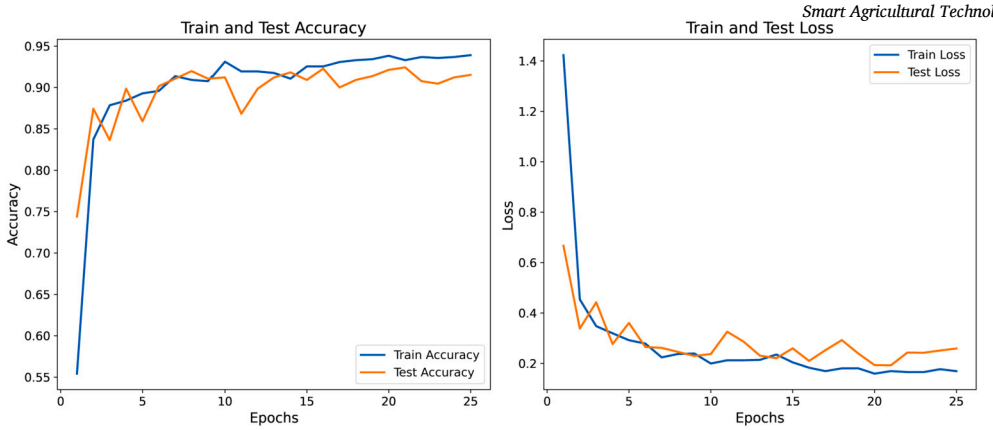


Fig. 18. Accuracy and loss vs. epochs of the applied CNN model.

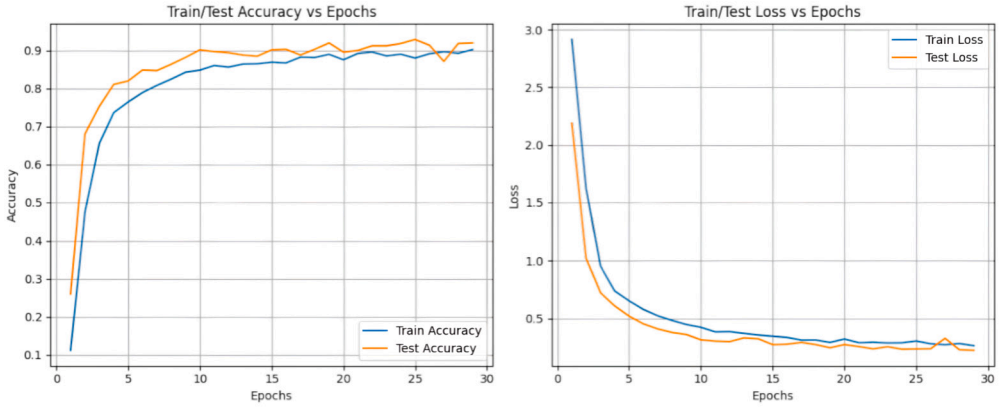


Fig. 19. Accuracy and loss vs. epochs of the applied LSTM model.

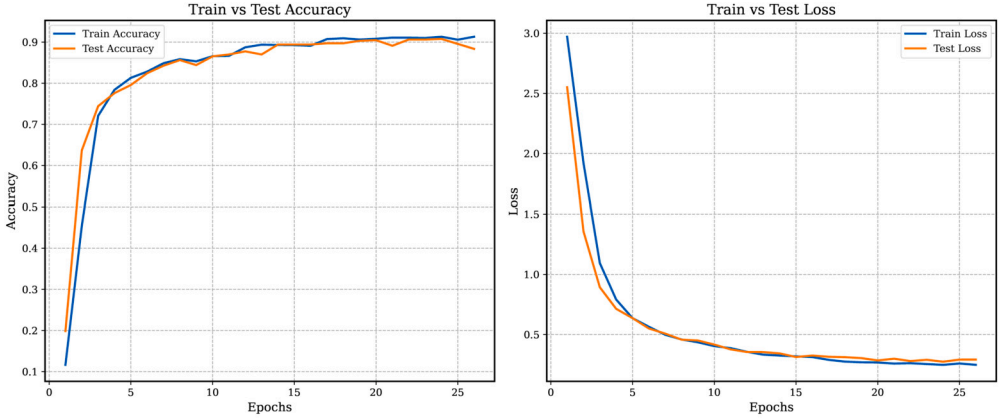


Fig. 20. Accuracy and loss vs. epochs of the applied CNN-LSTM model.

lizes at approximately 90%, with the testing accuracy closely following, indicating effective learning. Both training and testing loss decrease rapidly, reaching low values and suggesting model convergence. The slight difference between training and testing accuracies shows that overfitting is well-controlled, and the model generalizes well to unseen data.

Fig. 20 demonstrates the strong predictive capability of the CNN-LSTM model over 26 epochs. Both training and testing accuracies show a steady increase, converging at approximately 89%. Concurrently, the corresponding loss values decrease and stabilize, indicating effective learning with minimal overfitting.

Fig. 21 presents a comparative analysis of training and testing performance across 25 epochs using dual visualizations. The left plot displays accuracy trends, while the right plot shows loss trends. Both training and testing accuracies exhibit a sharp initial increase, stabilizing at approximately 90%. The loss plot demonstrates a rapid decrease in both training and testing loss during the early epochs, converging after around 10 epochs. The stable accuracy and loss curves indicate a robust model effectively converging to high performance on both datasets.

As previously observed, TabNet achieved the highest accuracy (92%) among the evaluated deep learning models. The ROC curve in Fig. 22 further illustrates this superior performance, which demonstrates a near-perfect AUC of 1.00 for all classes. This result signifies excellent classification performance and strong discriminative power between classes.

The confusion matrix in Fig. 23 demonstrates the applied TabNet model's generally robust performance across various crop types. The

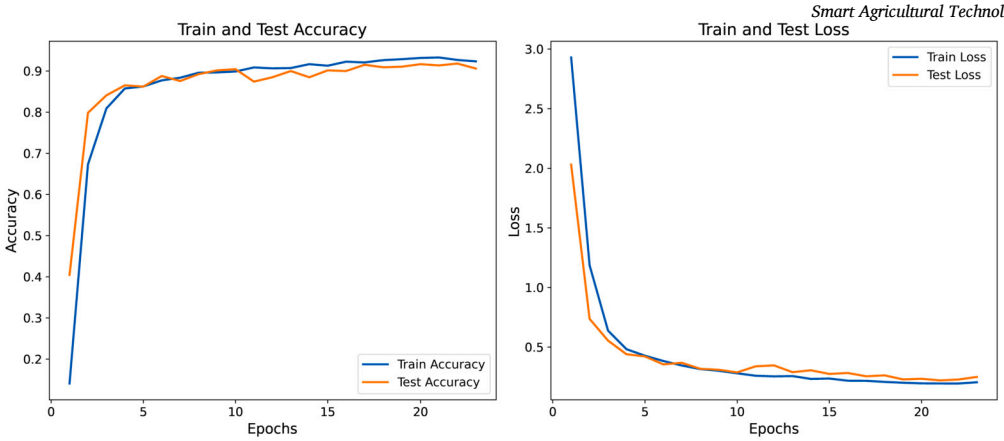


Fig. 21. Accuracy and loss vs. epochs of the applied CNN-BiLSTM model.

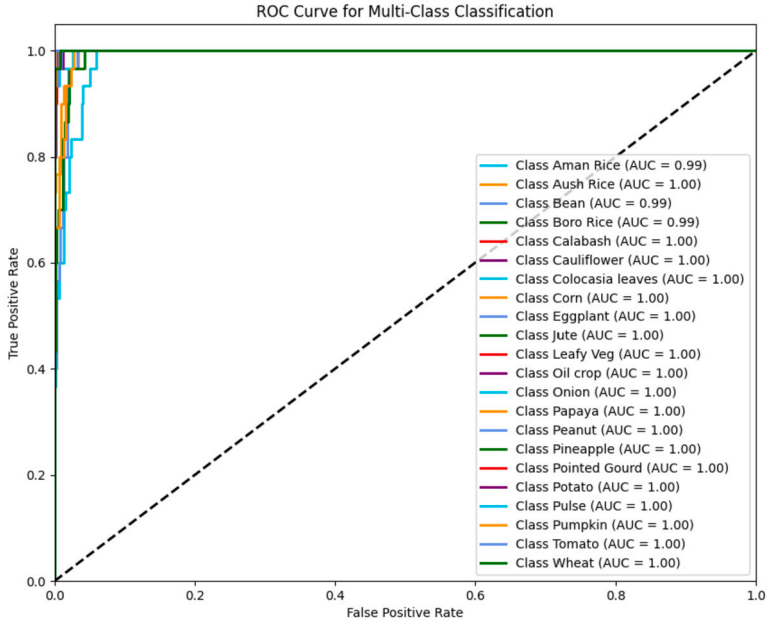


Fig. 22. ROC curve of the applied TabNet model.

strong diagonal values indicate high accuracy and low error rates for most crops, including Boro Rice, Cauliflower, Oil Crop, and Tomato. However, some misclassifications are evident, particularly between Aman Rice and Aush Rice, which exhibit a substantial number of misclassified samples.

Fig. 24 visualizes the LIME explanation for a sample test instance using the Gradient Boosting model. The model correctly predicts the crop with high probability, leveraging the following key features: Nitrogen, Phosphorus, Rainfall, pH, Potassium, Temperature, and Humidity. Rainfall ($0.26 < \text{Rainfall} \leq 0.41$) and Potassium ($K \leq 0.07$) contribute positively to the prediction, while high Humidity ($\text{Humidity} > 0.67$) has a negative influence.

Table 7 compares the proposed study with recent works conducted in precision agriculture and IoT-based crop recommendation systems. The best model in the proposed study is the SVC meta learner-based stacking ensemble, which achieves an accuracy of 95.9% and an F1-score of 0.958. Among the related works, the highest accuracy of 99% was achieved by Ridge Regression with the dataset of 8 features and 1400 samples.

Table 8 presents the performance metrics of various models deployed in the proposed IoT device, comparing accuracy and inference time. The Stacking ensemble technique with the SVC meta learner attained the highest accuracy of 95.9% with an inference time of 0.4347 seconds. Random Forest and Gradient Boosting also performed well, achieving accuracies of 95.8% and 95.5%, with inference times of 0.0608 s and 0.0693 s, respectively.

The proposed real-time soil nutrient-based IoT system for crop suggestions and field monitoring in Bangladesh was evaluated through a survey conducted in January 2025. The survey included 10 participants, comprising farmers, agricultural researchers, and agriculture students (8 males and 2 females). The first six questions assessed usability, interface clarity, real-time data satisfaction, accuracy of crop suggestions, system responsiveness, and reliability, with average ratings ranging from 3.78 to 4.22 on a 5-point scale. The last six questions were subjective, gathering qualitative feedback on desired features, improvements, decision-making utility, real-world applications, recommendation likelihood, and effectiveness in addressing user needs. Fig. 25 presents the average ratings from user feedback on six key aspects of the IoT-based soil nutrient monitoring system, where “Interface Clarity” and “Responsiveness” received the highest ratings (4.22). In contrast, “Accuracy” received the lowest (3.78), indicating areas for potential improvement.

Fig. 26 illustrates key terms from the responses to the five subjective survey questions, highlighting frequently mentioned concepts such as “system,” “useful,” “crop,” “farmers,” “data,” and “accuracy.” The word cloud-based visualization indicates that users emphasized the system’s

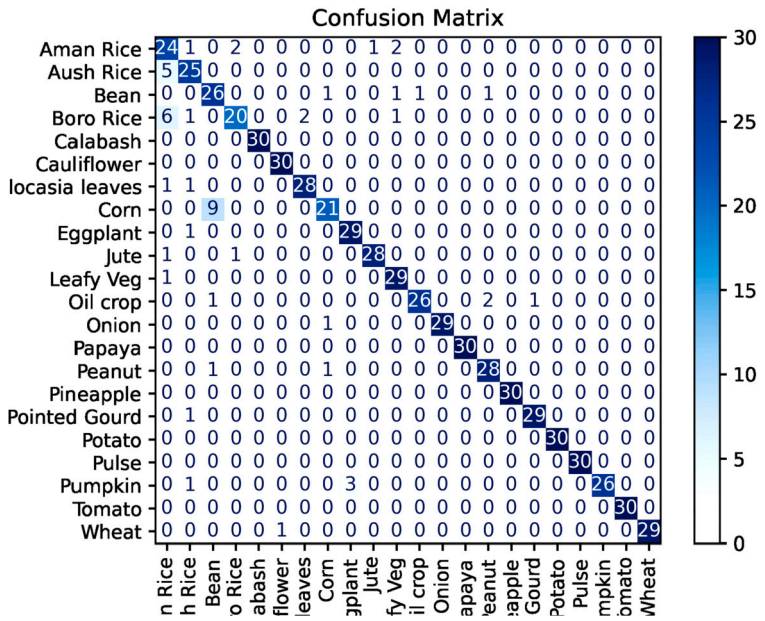


Fig. 23. Confusion matrix of the TabNet model.

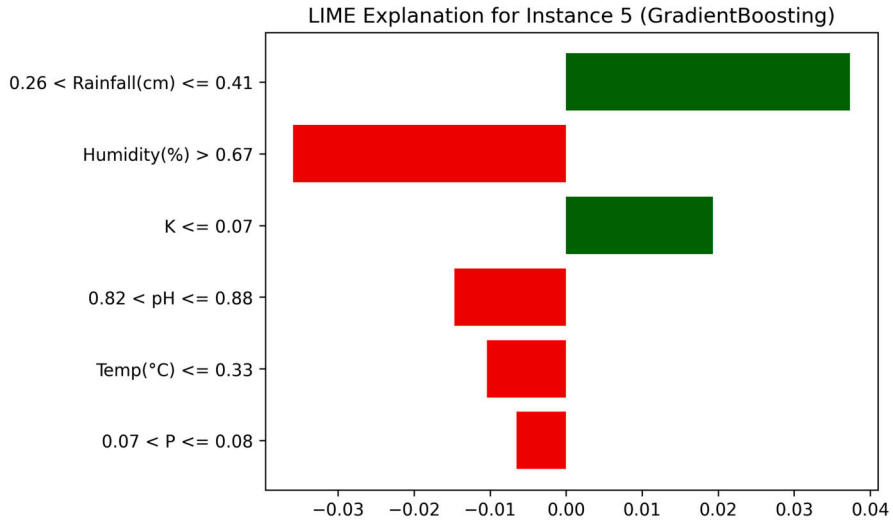


Fig. 24. Visualization of various features influenced by LIME.

Table 7
Comparison of the Proposed Crop Prediction System with Related Works.

Ref.	Dataset	Best Model	Accuracy	Metrics
[20]	Class: 22, Features: 7, Sample: 2299	CatBoost	97.5%	F1-score = 0.975
[22]	Class: 5, Features: 5, Sample: 645	XAI-Crop	94.15%	MSE = 0.941
[23]	Class: 5, Features: 8, Sample: 1400	Ridge Regression	99%	MSE = 0.920
[24]	Class: 20, Features: 7, Sample: 2490	GradBoost	98.17%	F1-score = 0.956
[25]	Class: 22, Features: 7, Sample: 2200	Decision Tree	92.17%	F1-score = 0.971
[26]	Class: 8, Features: 4, Sample: 10440	GRNN	97%	F1-score = 0.890
[2]	Class: 20, Features: 2, Sample: N/A	SVM	97.3%	F1-score = 0.932
[27]	Class: 22, Features: 8, Sample: 2199	Random Forest	97%	F1-score = 0.960
[28]	Class: 22, Features: 8, Sample: 2600	Random Forest	92%	F1-score = 0.912
[29]	Class: 4, Features: 7, Sample: 100	KNN	84%	F1-score = 0.836
[30]	Class: 22, Features: 8, Sample: 2300	Random Forest	99.54%	F1-score = 0.992
[31]	Class: 16, Features: 7, Sample: 1600	Random Forest	99.31%	F1-score = 0.991
[11]	Class: 18, Features: 5, Sample: 1950	Random Forest	99.31%	F1-score = 0.989
Our Study	Class: 22, Features: 8, Sample: 3300	SVC meta learner-based Stacking ensemble	95.9%	F1-score = 0.958

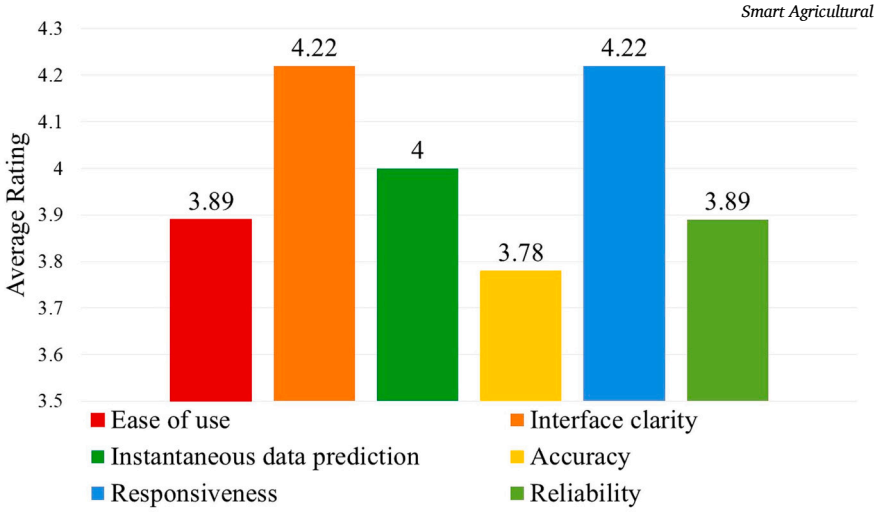


Fig. 25. User evaluation of the proposed IoT-based crop prediction system.

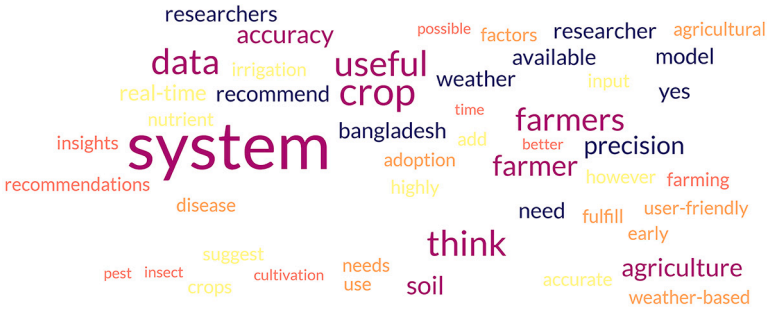


Fig. 26. Word cloud representation of user feedback on the proposed device.

Table 8
Performance Metrics of Various Models Deployed in the Proposed IoT Device.

Model	Accuracy	Inference Time
Stacking (SVC meta learner)	95.9%	0.4347 s
Random Forest	95.8%	0.0608 s
Gradient Boosting	95.5%	0.0693 s

utility, precision, and potential for adoption while also identifying areas for improvement, i.e., real-time data integration, weather-based recommendations, and user-friendliness.

4.1. Challenges and limitations

Despite the effectiveness, the proposed edge-AI crop-recommendation system faces several practical and structural constraints that currently hinder their broad deployment, while simultaneously highlighting fertile research directions in agronomic AI, edge computing, and sustainable farming.

Geographic generalizability: The present model is trained exclusively on soil, climate, and crop observations from Habiganj District (Sylhet Division), encompassing 22 widely cultivated crops. Consequently, it may underperform in ecotypes characterizing Bangladesh's saline coastal belt, drought-prone uplands, or annual floodplains, where edaphic and micro-climatic regimes differ substantially.

Heterogeneous data integration: Scaling nationwide demands the fusion of region-specific datasets, whose labeling conventions, feature ranges, and temporal resolutions are rarely harmonized. Achieving schema alignment without inflating on-device inference latency remains an open engineering problem.

Crop-diversity bias: The present focus on high-acreage staples risks marginalizing climate-resilient or regionally adapted varieties—e.g., Kalijira rice and local pulses—that are ecologically valuable yet sparsely represented. Collecting balanced, high-quality samples for such niche crops is costly and may exacerbate class imbalance.

Micronutrient-sensing latency: Critical nutrients, such as sulfur and zinc, are still quantified via delayed laboratory assays, introducing decision-making lags and limiting field-level granularity. Embedding low-cost, low-power sensors—such as portable X-ray fluorescence (pXRF) and ion-selective electrodes—into smallholder settings that meet the form-factor, cost, and power budgets is an unsolved challenge.

Meteorological resolution and volatility: Historical BAMIS records and coarse API feeds do not capture intra-seasonal anomalies, such as flash droughts and untimely precipitation. This limits recommendations in increasingly unpredictable climate conditions. We must balance the integration of near-real-time, high-resolution products, such as CHIRPS and ERA5-Land, or ensemble forecasts, with the computational limitations of low-resource devices.

External validation and robustness: Model assessment has thus far relied on internal cross-validation, leaving generalization across agro-ecological zones unverified. Federated or domain-adaptive learning approaches—which preserve data sovereignty while enabling both horizontally and vertically distributed training—will be essential for trustworthy, region-agnostic deployment.

Future Outlook: Overcoming these obstacles is essential for transforming the prototype into a scalable, equitable, and climate-resilient decision-support system. Attaining this objective will require strong interdisciplinary collaboration across fields such as agronomy, embedded systems, remote sensing, and machine learning to develop models that are not only technically sound but also socially responsible and environmentally conscious.

5. Conclusion

This study assesses the ability of classical algorithms and modern deep-learning techniques to raise crop-type prediction accuracy on resource-constrained edge devices, thereby advancing precision agriculture. The experimental dataset comprises 3,300 instances representing 22 major crops, each annotated with eight edaphic and agrometeorological variables gathered across eight unions in Habiganj District, Sylhet, Bangladesh, using datasets from the Soil Research Development Institute (SRDI) and the Bangladesh Agri-Meteorological Information Service (BAMIS).

Among the classical models, Random Forest, Gradient Boosting, and a stacked SVC ensemble achieved top accuracies of 95.8%, 95.5%, and 95.9%, respectively. Owing to its superior inference latency, Random Forest was deployed on a Raspberry Pi 5-based IoT platform for real-time recommendations. TabNet delivered the strongest performance within the deep-learning cohort, attaining 92% accuracy.

Model interpretability improved with LIME, and it consistently identified nitrogen, phosphorus, and rainfall as the main predictive factors. A mixed-methods survey of farmers and agricultural scientists confirmed the prototype's usefulness and pointed out areas for improvement. Consequently, the hardware device is a decision-support tool that supplies data-driven insights critical to sustainable cropping and global food-security initiatives.

Researchers and practitioners are encouraged to adopt the proposed IoT architecture for ongoing agri-environmental monitoring. Its rich sensor streams can readily integrate with federated and distributed learning, edge computing, advanced model compression, and AI-driven analytics, further elevating predictive fidelity. Nationwide deployment in Bangladesh would create a real-time data framework that adaptive AI models could easily use. This would improve scalability and accuracy. Future work will expand the framework to include datasets related to pests and diseases, various soil types, and detailed nutrient profiles, allowing edge devices to diagnose in-field diseases effectively using advanced machine-learning and deep-learning methods.

CRediT authorship contribution statement

MD Shaifullah Sharafat: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Nilavro Das Kabya:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Rahimul Islam Emu:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation. **Mehrab Uddin Ahmed:** Writing – original draft, Visualization, Validation, Methodology. **Jakaria Chowdhury Onik:** Writing – original draft, Visualization, Validation, Data curation. **Mohammad Aminul Islam:** Writing – review & editing, Project administration, Funding acquisition. **Riasat Khan:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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